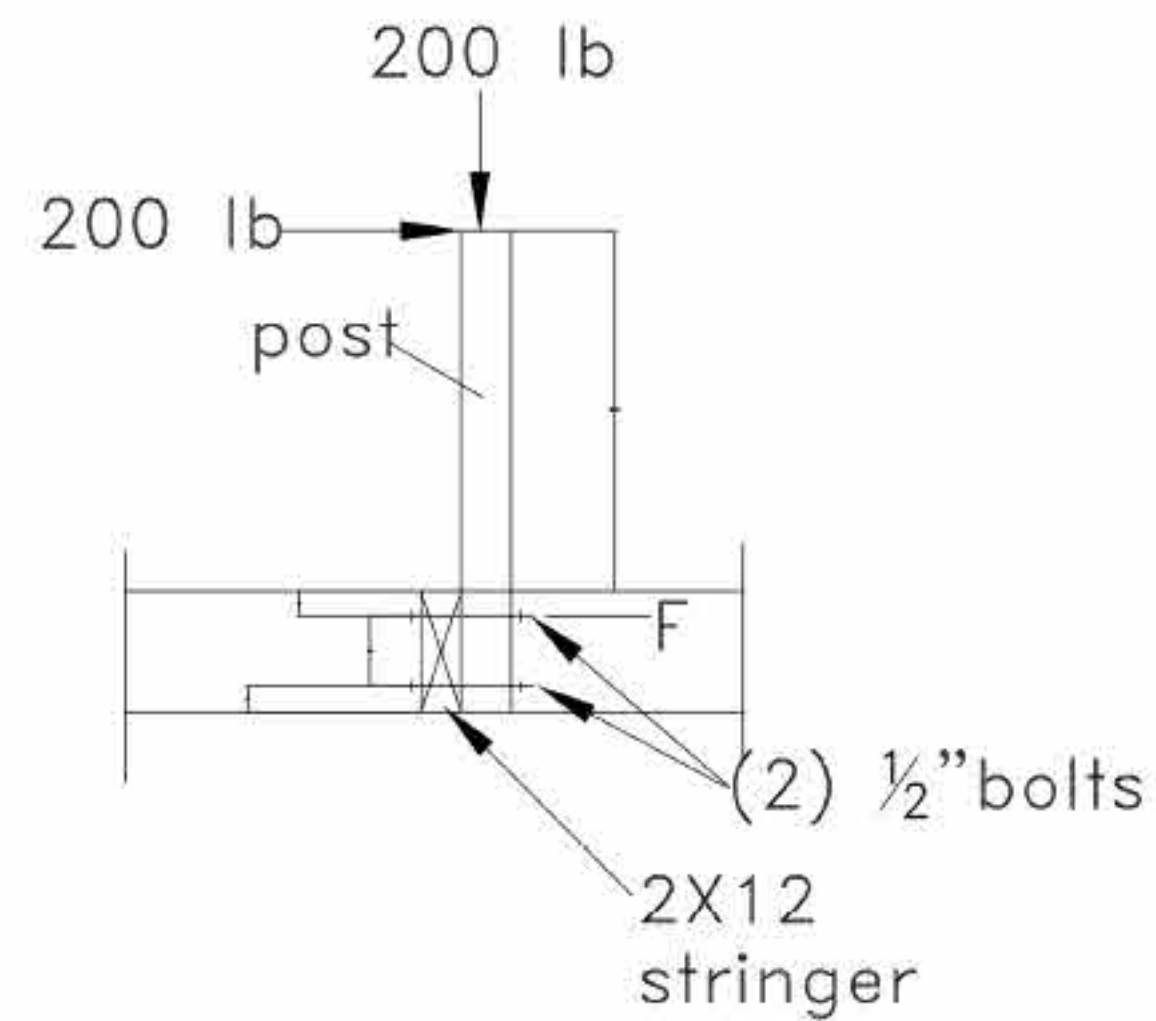




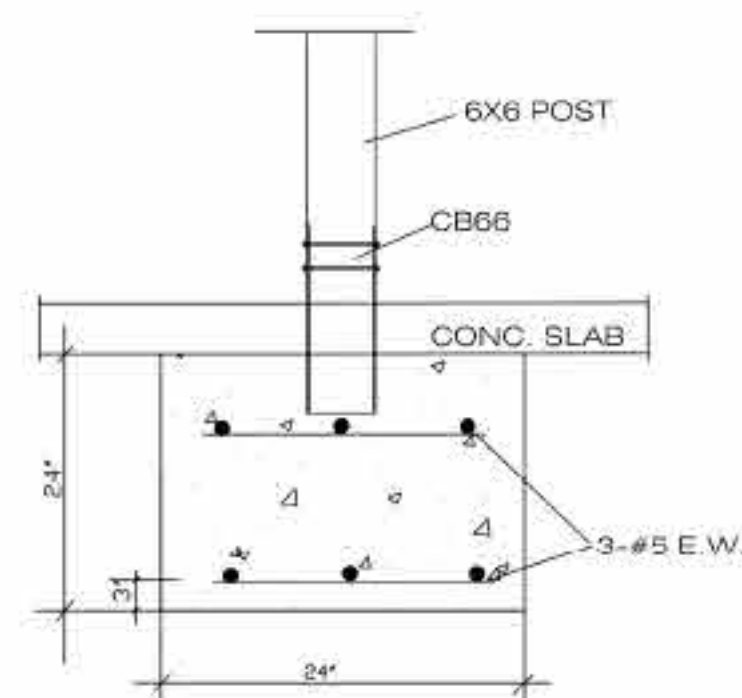
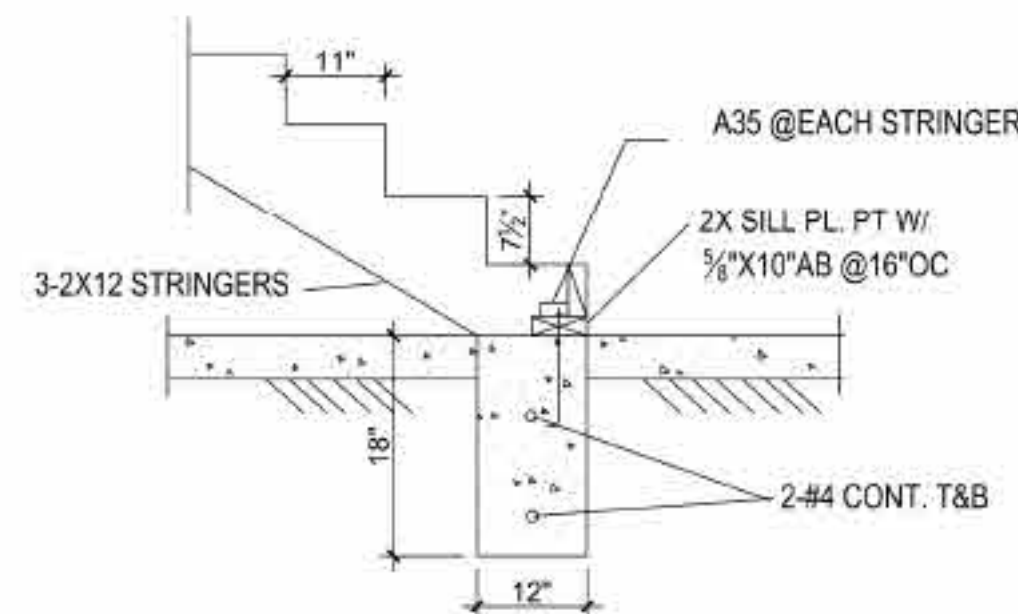
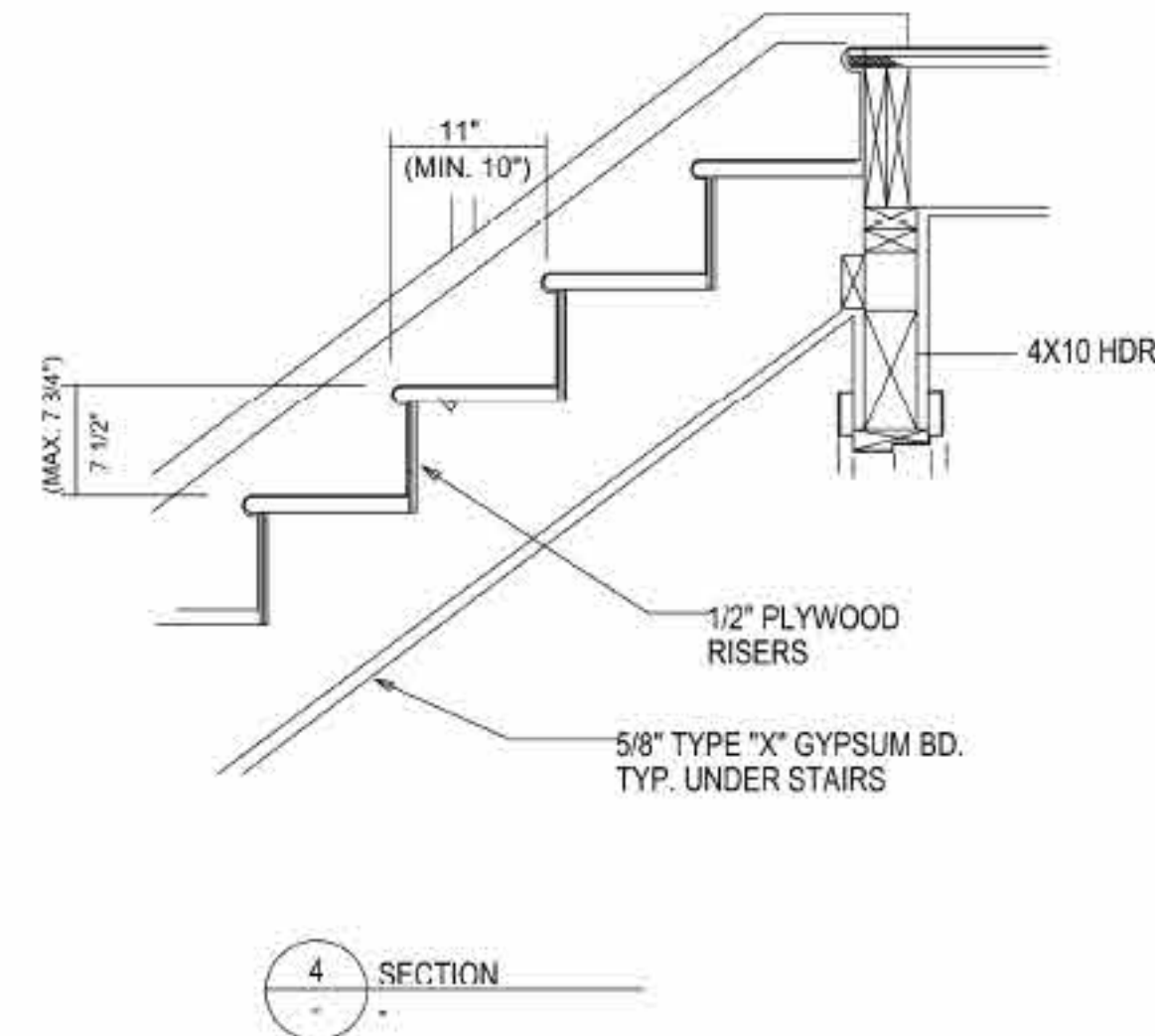
HANDRAIL CALS

P=200 lb (HORIZONTAL)
 $M = 200 \times 48" = 9600 \text{ IN-LB}$
 $F = 9600 / 7.5" = 1280 \text{ LB}$
 USE TWO 1/2" BOLTS A307
 ALLOW $F_t = 3900 \text{ lb}$
 P=200 lb (VERTICAL AT POST)
 ALLOW SHEAR = 550 lb (each bolt)

P=200 lb (VERTICAL AT CENTER SPAN)
 $w = 200 / 4 = 50 \text{ plf}$
 $M = (50 \times 4^2) / 8 = 100 \text{ ft-lb}$
 $R = 50 \times 4 / 2 = 100 \text{ lb}$
 $2 \times 4 \quad A = 5.25 \quad S = 3.06$
 $f_b = 100 \times 12 / 3.06 = 392 \text{ psi}$
 $f_v = 1.5 \times 100 / 5.25 = 8.6 \text{ psi}$

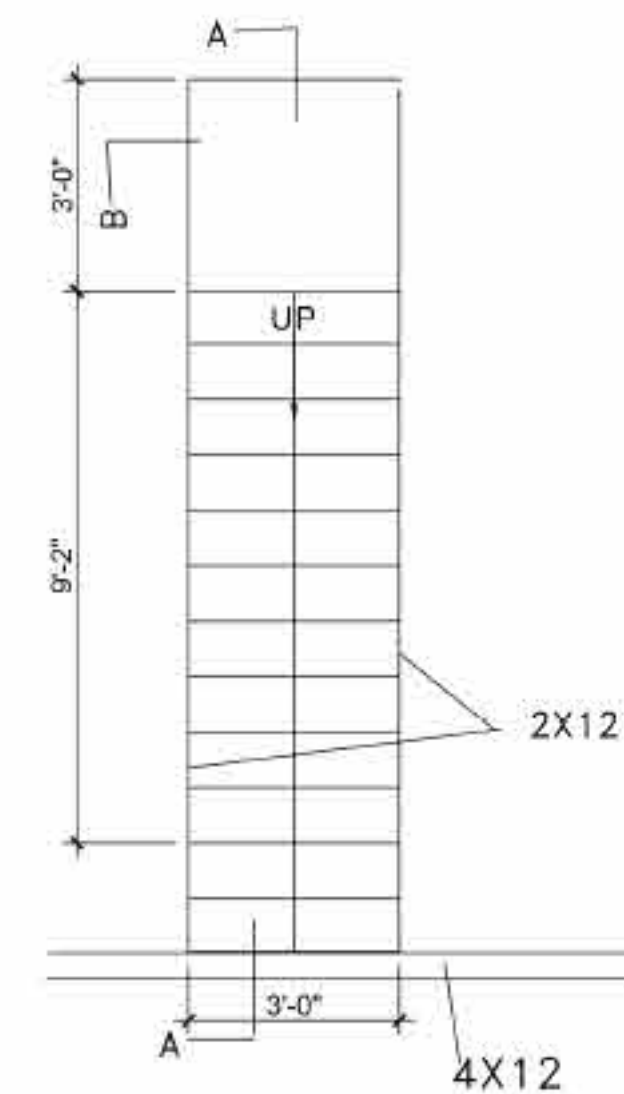
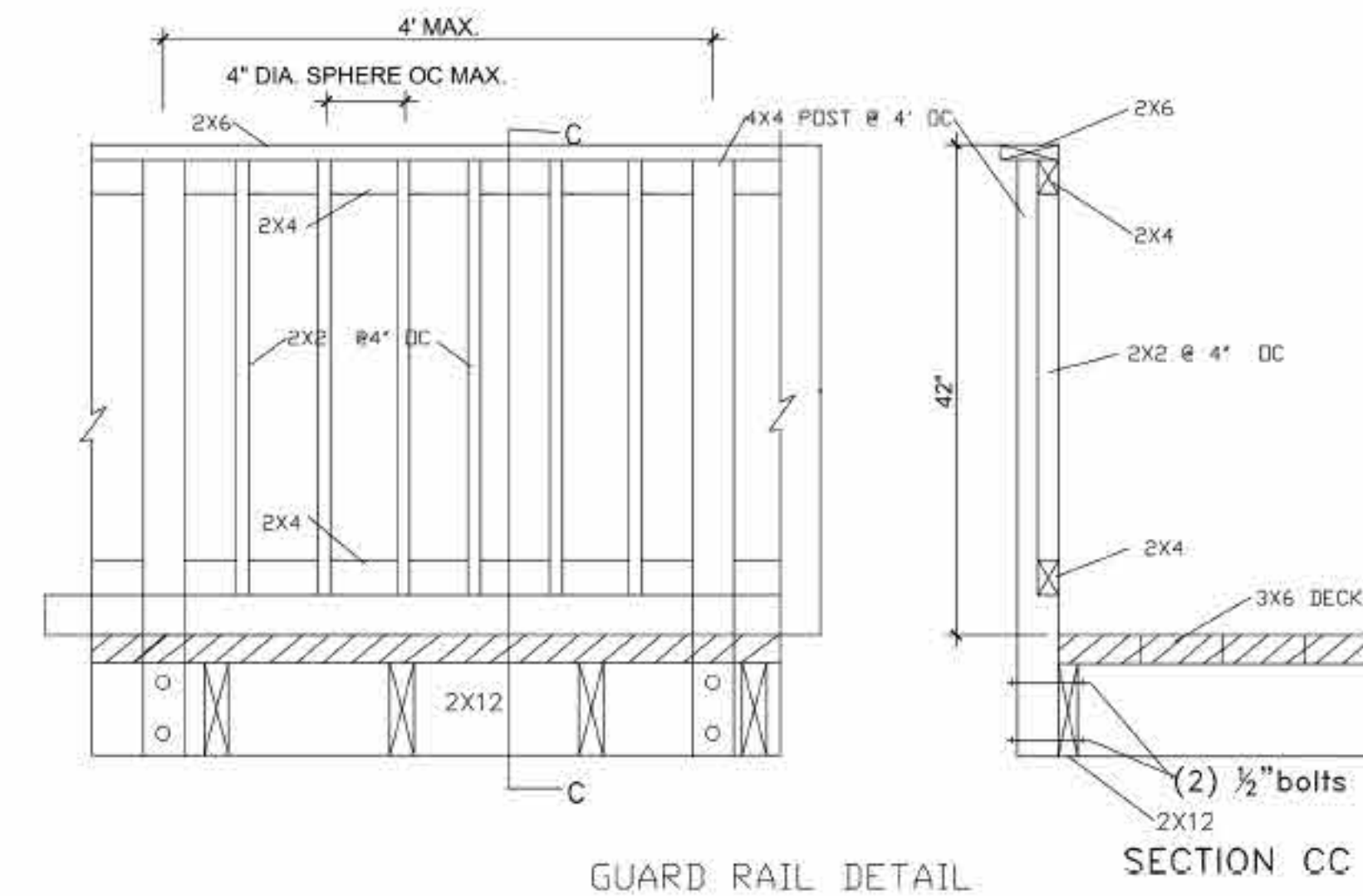
STAIR CALS

$L = 9'-2"$
 $DL = 12 \text{ PSFX} 11.1 / 9.2 = 14.5 \text{ PSF}$
 $LL = 40 \text{ PSFX} 11.1 / 9.2 = 48.3 \text{ PSF}$
 $w = (14.5 + 48.3) \times 3.5 / 2 = 110 \text{ plf}$
 $M = 110 \times 9.2^2 / 8 = 1164 \text{ ft-lb}$
 $R = 110 \times 9.2 / 2 = 506 \text{ LB}$
 USE MINIMUM TWO 2X12 STRINGERS
 $S = 31.64 \quad A = 16.88 \quad I = 178$
 $DF-L \text{ NO. 2} \quad F_b = 900 \text{ psi} \quad F_v = 180 \text{ psi} \quad E = 1600 \text{ ksi}$
 $f_b = 1164 \times 12 / 31.64 = 441.5 \text{ psi}$
 $f_v = 1.5 \times 506 / 16.88 = 45 \text{ psi}$
 $DEFL = 5w^4 / 384EI = 0.01"$

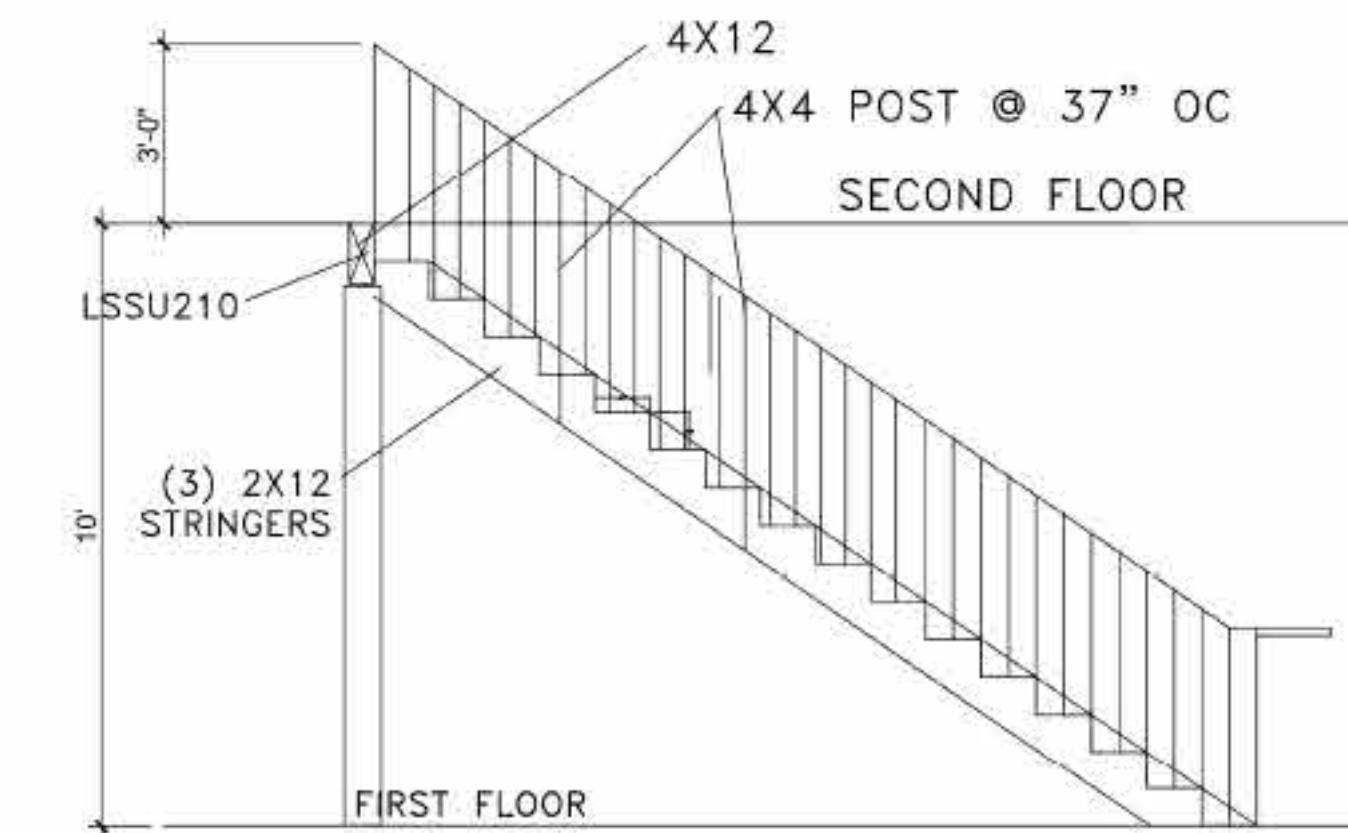


FTG @ POST

2 COLUMN TO FTG



STAIR FRAMING PLAN



SECTION AA

3 STAIR FRAMING DETAILS